

Full Notes

Fundamentals of Engineering Materials

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1 Introduction

1.1 Materials Science and Engineering

Definition: Materials Science

Investigates relationships between structures and properties of materials, and designs/develops new materials. Source: 2.PNG

Definition: Materials Engineering

Creates products from existing materials and develops materials processing techniques. Source: 2.PNG

Definition: Core terms

- **Composition:** chemical make-up of a material.
- **Structure:** arrangement of atoms in a material.
- **Processing:** ways to shape materials into useful components.

Source: 2.PNG

The materials chain

Composition → Structure → Processing → Properties → Performance

Source: 2.PNG

Definition: Materials knowledge spectrum

Materials science provides *basic* knowledge; materials engineering provides *applied* knowledge. Together they produce the resultant knowledge of structure, properties, processing and performance that engineers use to convert materials into the products needed by society. Source: 6.PNG

Example: Structure, not composition, sets properties

A disk of aluminum oxide in three forms (single crystal, polycrystalline, and polycrystalline with many pores) has different optical properties in each form despite identical composition. Source: 3.PNG

1.2 Why Materials Are Important

Definition: Why it matters

Materials drive advancements in society. Engineers must understand materials in order to select appropriate materials and processing techniques for specific applications. Source: 4.PNG, 5.PNG

Definition: Historical materials ages

- Stone Age, Bronze Age, Iron Age mark society's advancement.
- Today's age is posed as an open question. The candidates offered are the Silicon (Electronic Materials) Age, the Nano-materials Age, and the Polymer Age.

Source: 4.PNG

Definition: Engineer's requirement

Products, devices and components are all made of materials. Selecting appropriate materials and processing requires knowledge of material properties and of structure–property relationships.

Source: 5.PNG

1.3 Types of Materials

Definition: Three broad classes

Metals and alloys; ceramics and glasses; polymers and plastics. Source: 7.PNG

1.3.1 Metals and Alloys

Definition: Alloy

A metal that contains additions of one or more metals or non-metals. Source: 8.PNG

Definition: Metals: structure and properties

- Pure metallic elements or combinations of them; a large number of de-localized electrons, and therefore able to conduct electricity. Source: 7.PNG
- Properties: good thermal and electrical conductivity, high strength, high stiffness, ductility. Source: 8.PNG
- Steel is an alloy of iron and a small content of carbon. Brass is an alloy of copper and zinc. Source: 8.PNG
- Examples: aluminum, magnesium, zinc, iron, titanium, beryllium-copper, nickel, copper-nickel, niobium. Source: 8.PNG

1.3.2 Ceramics and Glasses

Definition: Ceramics

Compounds of metallic and non-metallic elements: oxides, carbides, nitrides, sulfides; generally insulating and refractory. Source: 7.PNG Inorganic non-metallic materials, crystalline (polycrystalline) in structure; examples include sand and rocks. Properties: insulating (non-conductive), high hardness, high strength and melting point, very brittle. Source: 9.PNG

Definition: Glasses

Amorphous materials with structure-dependent properties, which can be processed to yield an improvement in properties. Source: 9.PNG

1.3.3 Polymers and Plastics

Definition: Polymer

“Poly” means many, “mer” means unit. Polymers are large molecular structures formed from monomers through polymerization. Source: 10.PNG

Definition: Polymers: composition and properties

- Chemical composition including carbon, hydrogen and other non-metals; large molecular structure; flexible, lightweight, easily processed. Source: 7.PNG
- Synthetic: polyethylene (PE), polymethyl methacrylate (PMMA), polycarbonate. Natural: cellulose, glucan. Source: 10.PNG
- **Practice.** Classify as metal, ceramic or polymer: brass, MgO, plexiglass, polychloroprene, B₄C, cast iron. The source provides no answer key. Source: 14.PNG

1.4 Other Materials

Definition: Composite

A combination of two or more different materials, combining the properties of the materials it is formed from. Source: 11.PNG, 12.PNG

Definition: Advanced materials

Semiconductors, biomaterials, smart materials, and nanoengineered materials. Source: 11.PNG

Definition: Semiconductors

Conductivity in between ceramics and metals. Used in computers, electronics and micro-electrical-mechanical systems (MEMS). Examples: silicon, germanium, gallium arsenide. Source: 13.PNG

1.5 Comparing Material Properties

Definition: Material property chart

Compares classes of materials (metals, ceramics, polymers, composites) against measurable properties such as density, elastic modulus and tensile strength, often on logarithmic scales. Source: 15.PNG

Ranking by class

Property	Highest	Lowest
Density	metals	polymers
Elastic modulus	metals, ceramics	polymers
Tensile strength	composites, metals, ceramics	none

Composites and woods occupy a wide middle range for density and stiffness. For tensile strength every class spans a range, so no class is lowest outright. Source: 15.PNG, 16.PNG, 17.PNG

Definition: Ashby map (two-property optimization)

Plots stiffness (elastic/Young's modulus, GPa) against density (kg/m³) on logarithmic axes, grouping materials into regions for technical ceramics, non-technical ceramics, composites, metals, natural materials, polymers, foams and elastomers. Used to compare and select materials on two properties simultaneously. Source: 18.PNG

1.6 Materials Selection Process

Method: Selecting a material and process

1. For a specific application, determine the required properties: mechanical, electrical, thermal, magnetic, optical, deteriorative.
2. From the list of required properties, identify candidate material(s).
3. For the best candidate material(s), specify processing technique(s), to provide the required set of properties and to produce a component of the desired shape and size. Example techniques: casting, mechanical forming, welding, heat treating.

Source: 19.PNG

2 Atomic Structure and Interatomic Bonding

Definition: Source keys

- **Pre-lecture:** before lecture-Chapter 2 - Atomic Structure and Interatomic Bonding.pdf
- **Full deck:** F26-D2L-Chapter 2- Introduction to Engineering Materials-Part 1 and part 2 (full).pdf
- **Week 2 problems:** EXAMPLE PROBLEMS 1 - WEEK 2.docx

2.1 Atomic Structure

Definition: Particles

Electrons (9.11×10^{-31} kg), protons and neutrons (each $\approx 1.67 \times 10^{-27}$ kg). Source: Pre-lecture, slide 3

Definition: Atomic number

The number of protons in the nucleus, equal to the number of electrons in a neutral species. Source: Pre-lecture, slide 3

Definition: Quantum numbers

- n , principal (shell): K, L, M, N, O for 1, 2, 3, 4, ...
- ℓ , azimuthal (subshell): s, p, d, f for 0, 1, 2, 3, ..., $n - 1$
- m_ℓ , magnetic (number of orbitals): 1, 3, 5, 7, ranging $-\ell$ to $+\ell$
- m_s , spin: $\frac{1}{2}$, $-\frac{1}{2}$

Source: Pre-lecture, slide 5

Definition: Maximum electrons per subshell

s = 2, p = 6, d = 10, f = 14. Source: Full deck, slide 22

Definition: Valence electrons

Electrons in the outer unfilled shells. They are available for bonding and tend to determine chemical properties. Filled shells are more stable and require more energy to gain or lose electrons. For most elements the configuration is not stable, because the valence shell is usually not completely filled. Source: Pre-lecture, slide 8; Full deck, slide 23

Method: Writing an electron configuration (Aufbau)

1. Fill orbitals bottom-up, so electrons occupy the lowest available energy states.
2. Respect the per-subshell maxima: s 2, p 6, d 10, f 14.
3. Identify the valence electrons as those in the outermost unfilled shell.

Source: Full deck, slides 20–22

Example: Nitrogen

Bottom-up filling gives $1s^2 2s^2 2p^3$, also written $[\text{He}] 2s^2 2p^3$. Source: Full deck, slide 22

Definition: Reference configurations

- Stable: He $1s^2$; Ne $1s^2 2s^2 2p^6$; Ar $1s^2 2s^2 2p^6 3s^2 3p^6$; Kr $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
- Others: H $1s^1$; Li $1s^2 2s^1$; Be $1s^2 2s^2$; B $1s^2 2s^2 2p^1$; Na $[\text{Ne}] 3s^1$; Mg $[\text{Ne}] 3s^2$; Al $[\text{Ne}] 3s^2 3p^1$
- C (6): $1s^2 2s^2 2p^2$, valence $2s^2 2p^2$
- Fe (26): $1s^2 2s^2 2p^2 3s^2 3p^6 3d^6 4s^2$, valence $3d^6 4s^2$

Source: Pre-lecture, slides 7–9; Full deck, slide 25

Definition: Wave and particle character

Electrons have both. Particle-like: an electron exists in a point orbiting a nucleus. Wave-like: position in an orbital is a probability distribution, and each orbital has a discrete energy level set by quantum numbers. Source: Full deck, slide 18

2.2 Atomic Mass, Atomic Weight, and the Mole

Definition: Approximate particle masses

Proton ≈ 1 amu, neutron ≈ 1 amu, electron ≈ 0.00055 amu. $1 \text{ amu} \approx 1.67 \times 10^{-24} \text{ g}$
 $= 1.67 \times 10^{-27} \text{ kg}$. Source: Full deck, slide 4

Definition: The amu

$1 \text{ amu} = \frac{1}{12}$ the mass of one carbon-12 atom, so one carbon-12 atom = 12 amu. Also written u. Grams are far too large for one atom, since a carbon-12 atom weighs only $\approx 1.99 \times 10^{-23} \text{ g}$. Source: Full deck, slide 9

The amu–gram bridge

$$1 \text{ amu} = \frac{1 \text{ g}}{6.022 \times 10^{23}} \approx 1.66 \times 10^{-24} \text{ g} \quad 12 \frac{\text{amu}}{\text{atom}} \longleftrightarrow 12 \frac{\text{g}}{\text{mol}}$$

Source: Full deck, slide 8

Definition: The mole

A counting unit, like a dozen: $1 \text{ mol} = 6.022 \times 10^{23}$ particles (Avogadro's number), and 12 g of carbon-12 = 1 mol. A mole is a *number* unit like kilo or mega, so it carries no physical unit of its own. Source: Full deck, slide 5; Week 2 problems, Problem 4

Definition: Isotopes

Atoms of the same element with the same number of protons but a different number of neutrons. Mass number = protons + neutrons; atomic number = protons. Carbon: ^{12}C (6p, 6n, 98.93%), ^{13}C (6p, 7n, 1.07%), ^{14}C (6p, 8n, trace). Source: Full deck, slide 12

Atomic weight

$$\bar{A} = \sum (\text{isotope mass})(\text{fractional abundance})$$

Source: Full deck, slide 14

Example: Natural uranium

$Z = 92$, isotopes 99.274% ^{238}U , 0.720% ^{235}U , 0.006% ^{234}U . Using mass numbers as approximate masses,

$$\begin{aligned}\bar{A}_{\text{U}} &= (238)(0.99274) + (235)(0.00720) + (234)(0.00006) \\ &= 236.27212 + 1.69200 + 0.01404 = 237.97816 \text{ amu.}\end{aligned}$$

The published standard value is 238.02891 amu; the gap comes from using integer mass numbers instead of the actual isotope masses (238.051, 235.044, 234.041). Source: Full deck, slide 14; Week 2 problems, Problem 1

Weight versus mass

$$W = m g$$

Mass is the amount of matter (kg) and does not change; weight is the gravitational force (N) and does. A 60 kg person: on Earth $W = (60)(9.81) = 588.6 \approx 589 \text{ N}$; on the Moon $W = (60)(1.62) = 97.2 \approx 97 \text{ N}$. Source: Full deck, slide 6

Watch out

Mass and weight are not the same quantity. Mass (kg) is the amount of matter and stays the same, weight (N) is the gravitational force and changes with g . Source: Full deck, slide 6

Atomic mass and atomic weight are not interchangeable. Atomic mass refers to an individual atom, atomic weight is the weighted average over naturally occurring isotopes. Source: Full deck, slide 13

2.3 Periodic Table and Electronegativity

Definition: Electropositive and electronegative

Electropositive elements readily give up electrons to become positive ions. Group IA gives up 1, Group IIA gives up 2, the group left of the transition metals gives up 3. Electronegative elements readily acquire electrons to become negative ions, accepting 1 or 2 depending on group. Inert gases neither give up nor accept. Source: Pre-lecture, slide 10

Definition: Electronegativity

Ranges 0.7 to 4.0; large values indicate a tendency to acquire electrons. It decreases toward the lower-left of the periodic table (Fr = 0.7, Cs = 0.7) and increases toward the upper-right (F = 4.0, O = 3.5). Source: Pre-lecture, slide 11

Definition: Values given

H 2.1, Li 1.0, Be 1.5, C 2.5, N 3.0, O 3.5, F 4.0, Na 0.9, Mg 1.2, Cl 3.0, Fe 1.8, Cu 1.9. Source: Pre-lecture, slide 11

2.4 Bonding Forces and Energies

Net energy and net force

$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}, \quad n \approx 8$$

$$F_N = F_A + F_R, \quad F = -\frac{dE}{dr}, \quad E = \int -F dr$$

Source: Pre-lecture, slide 13; Full deck, slides 31, 44

The Coulombic constant A

$$A = \frac{1}{4\pi\epsilon_0} (|Z_1|e)(|Z_2|e)$$

with $\epsilon_0 = 8.85 \times 10^{-12}$ F/m the permittivity of vacuum, $|Z_1|, |Z_2|$ the absolute valences, and $e = 1.602 \times 10^{-19}$ C. This assumes a totally ionic bond; since most bonds are not 100% ionic, A is normally taken from experimental data rather than computed. Source: Full deck, slide 44

Definition: Equilibrium separation r_0

The separation at which E_N is a minimum (E_0) and $F_N = 0$. It is the bond length, the distance between the centers of two bonded atoms at lowest potential energy. For $r < r_0$ repulsion dominates and pushes atoms apart; for $r > r_0$ attraction dominates and pulls them together; as $r \rightarrow \infty$ both are negligible and $V \approx 0$. The equilibrium meant here is *mechanical*, not chemical, not thermal. Source: Pre-lecture, slide 13; Full deck, slides 32–35; Week 2 problems, Problem 3

Lennard-Jones potential

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

The r^{-12} term is the strong short-range repulsion; the negative r^{-6} term is the attraction. σ and ϵ are model parameters determined by experiment. Source: Full deck, slides 36–38; Week 2 problems, Problem 3

Definition: σ and ϵ

σ is the separation where the potential crosses zero, $V(\sigma) = 0$. ϵ is the depth of the potential well and is the bond energy, meaning the energy needed to separate two bonded atoms from equilibrium to infinity. A deeper well means stronger bonding. At equilibrium $V(r_0) = -\epsilon$. Source: Full deck, slides 34, 37, 38

Method: Finding r_0 and the bond energy from Lennard-Jones

1. Rewrite $V(r) = 4\epsilon[\sigma^{12}r^{-12} - \sigma^6r^{-6}]$ and differentiate: $\frac{dV}{dr} = 4\epsilon \left[-\frac{12\sigma^{12}}{r^{13}} + \frac{6\sigma^6}{r^7} \right]$.
2. Set $dV/dr = 0$ at $r = r_0$. Since $\epsilon > 0$, divide by 4ϵ : $\frac{12\sigma^{12}}{r_0^{13}} = \frac{6\sigma^6}{r_0^7}$.
3. Multiply by r_0^{13} : $12\sigma^{12} = 6\sigma^6r_0^6$, so $r_0^6 = 2\sigma^6$ and $r_0 = \sqrt[6]{2}\sigma \approx 1.122\sigma$.
4. Substitute back. With $(\sigma/r_0)^6 = \frac{1}{2}$ and $(\sigma/r_0)^{12} = \frac{1}{4}$: $E(r_0) = 4\epsilon \left[\frac{1}{4} - \frac{1}{2} \right] = -\epsilon$.

Source: Full deck, slides 40–42; Week 2 problems, Problem 3

Example: Force between K^+ and O^{2-}

Centers separated by 1.5 nm. The attractive force is the negative derivative of the attractive energy with respect to separation, using $A = \frac{1}{4\pi\epsilon_0}(|Z_1|e)(|Z_2|e)$. Valences +1 and -2 give $Z_1 = 1$, $Z_2 = 2$, and a magnitude of 2.05×10^{-10} N, reported without sign, since the force is known to be attractive. Source: Full deck, slide 45; Week 2 problems, Problem 5

Bonding energy and melting temperature

The larger the bonding energy E_0 (deeper well), the higher the melting temperature T_m .

Bond	Material	kJ/mol	T_m (°C)
Ionic	NaCl	640	801
	LiF	850	848
	MgO	1000	2800
	CaF ₂	1548	1418
Covalent	Cl ₂	121	-102
	Si	450	1410
	InSb	523	942
	C (diamond)	713	> 3550
	SiC	1230	2830
Metallic	Hg	62	-39
	Al	330	660
	Ag	285	962
	W	850	3414
van der Waals	Ar	7.7	-189
	Kr	11.7	-158
	CH ₄	18	-182
	Cl ₂	31	-101

Source: Pre-lecture, slide 29

Watch out

$\sigma \neq r_0$. σ is where the potential crosses zero; r_0 is where it reaches its minimum, with $r_0 = 2^{1/6}\sigma$. Source: Full deck, slides 34, 37

The well depth is $+\epsilon$ while the energy at the minimum is $-\epsilon$. Bond energy is quoted positive because it equals the positive work required to break the bond. Source: Full deck, slide 42; Week 2 problems, Problem 3

Source conflict. The pre-lecture deck writes the Lennard-Jones prefactor as 3ϵ ; the full deck and the Week 2 problems both use 4ϵ . Source: Pre-lecture, slide 30; Full deck, slides 36–38; Week 2 problems, Problem 3

2.5 Interatomic Bonding Overview

Definition: Primary and secondary

Primary: ionic, covalent, metallic. Secondary (van der Waals, including hydrogen bonding): from atomic or molecular dipole interactions. Secondary bonding exists between virtually all atoms or molecules but may be obscured when a primary type is present; it is weaker, yet still influences physical properties. Source: Pre-lecture, slide 14; Full deck, slide 62

Definition: Directional versus non-directional

- **Directional:** covalent and some secondary bonding. Electrons or shared electrons are localized between atoms; the bond exists only because of that location or dipole moment.
- **Non-directional:** ionic and metallic. Electrons cannot be localized; bonding forces are the same in all directions around the atom.

Source: Pre-lecture, slide 24

Relative strength of intramolecular forces

metallic > ionic > polar covalent > nonpolar covalent

Source: Pre-lecture, slide 27

Definition: Intramolecular versus intermolecular

Intramolecular forces hold atoms together within a molecule; intermolecular forces bond molecules together. Boiling and melting points depend on the type and strength of the intermolecular forces. Stronger forces require more energy to break, so ionic and polar covalent compounds have higher boiling and melting points. Source: Pre-lecture, slides 26, 28

Definition: Chapter summary

Chemical, electrical, thermal and optical properties are determined by electronic configuration. Materials with large bonding energies usually have high melting temperatures. The nature of the bond depends on the electron structure of the constituent atoms. Source: Full deck, slide 62

2.6 Ionic Bonding

Definition: Ionic bonding

Occurs between positive and negative ions. Requires electron transfer and a large difference in electronegativity. A metal atom donates electrons and becomes a stable cation; a non-metal accepts them and becomes a stable anion. The two are held together by Coulombic (electrostatic) attraction. Source: Pre-lecture, slides 15, 16

Example: MgO

Mg ([Ne]3s²) donates its 3s² electrons to O (1s²2s²2p⁴), forming Mg²⁺ ([Ne]) and O²⁻ ([Ne]).

Source: Pre-lecture, slide 16

Definition: Where it appears

The predominant bonding in ceramics. Examples: NaCl, MgO, CaF₂, CsCl. Source: Pre-lecture, slide 18

Method: Drawing a Lewis structure

1. Write the element symbol.
2. Determine the number of valence electrons from the element's group. chlorine is in group 7, so 7 valence electrons.
3. Draw dots for electrons around the symbol, starting at the right and going counter-clockwise, pairing as needed.

Source: Pre-lecture, slide 34

2.7 Covalent Bonding

Definition: Covalent bonding

Occurs between atoms with similar electronegativities, which share electrons. Bonds involve valence electrons, normally s and p orbitals. Source: Pre-lecture, slide 18

Definition: sp³ hybridization

Carbon can form sp³ hybrid orbitals: an electron is promoted from the 2s orbital into the 2p orbital, then hybridizes, producing four equivalent 2sp³ orbitals at 109.5° to one another. Source: Pre-lecture, slide 19

Example: CH₄

Carbon has 4 valence electrons and needs 4 more; each hydrogen has 1 and needs 1 more. Electronegativities of C and H are similar, so electrons are shared in sp³ hybrid covalent bonds, with orbital overlap between carbon's sp³ orbitals and each hydrogen's 1s orbital. Source: Pre-lecture, slide 20

2.8 Metallic Bonding

Definition: Metallic bonding

Electrons are delocalized to form an "electron cloud." Positively charged ion cores sit in a lattice surrounded by a sea of valence electrons delocalized across the whole structure rather than bound to individual atoms. It is the strongest of the intramolecular forces. Source: Pre-lecture, slides 21, 27

2.9 Secondary Bonding (Van der Waals)

Definition: Secondary bonding

Arises from attractive forces between dipoles.

- **Fluctuating dipoles:** from asymmetric electron clouds; liquid H₂ bonds between adjacent H₂ molecules this way.

- **Permanent dipoles:** a molecule with a $+$ end and a $-$ end bonds to another such molecule; liquid HCl, and chains in a linear polymer.

Source: Pre-lecture, slide 22

Definition: Hydrogen bonding

A special type of secondary bonding where hydrogen is one of the constituents: HF, H₂O, NH₃. Its magnitude is generally higher than other secondary bonds, and the melting and boiling temperatures of those compounds are correspondingly high. Source: Pre-lecture, slide 23

2.10 VSEPR Theory and Molecular Geometry

Definition: VSEPR

Valence Shell Electron Pair Repulsion: valence electron pairs around an atom repel each other, and greater repulsion means higher energy and less stability. VSEPR predicts the geometry that minimizes repulsion. Source: Pre-lecture, slide 32

Method: Predicting molecular geometry

1. Draw the Lewis structure.
2. Count electron groups (bonding pairs and lone pairs) around the central atom.
3. Determine the electron geometry.
4. Determine the molecular geometry from the number of bonding pairs versus lone pairs.

Source: Pre-lecture, slide 33

Definition: Notation AX_mE_n

A = central atom, X = bonding atom, E = non-bonded electrons, m = number of bonding atoms, n = number of non-bonded electron pairs. Source: Pre-lecture, slide 35

Shape by formula type

Type	Geometry	Bond angle
AX_2	planar	180°
AX_3	trigonal planar	120°
AX_4	tetrahedral	109.5°

Source: Pre-lecture, slide 36

Definition: Predicting distortions

Repulsion strength, greatest to weakest: lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair. Size, largest to smallest: lone pair > triple bond > double bond > single bond. Unpaired valence electrons take slightly more space than bonded electrons, which may shift bond angles; double and triple bonds repel more strongly than single bonds. Source: Pre-lecture, slides 38, 39